

Theorem (2.5) applies. In particular, $\text{Pic}^0(\mathcal{C}_s/k) = \mathcal{J}_s^0$, so $\text{Pic}^0(\mathcal{C}_s/k)$ has no unipotent radical. We apply:

Lemma (2.6). — *Let D be a complete 1-dimensional scheme over k such that $H^0(\mathcal{O}_D) = k$, and such that the generalized jacobian of D has no unipotent radical. Then*

- (i) $\chi(\mathcal{O}_D) = \chi(\mathcal{O}_{D_{\text{red}}})$;
- (ii) *the singularities of D_{red} are all transversal crossings of a set of non-singular branches (i.e., analytically isomorphic to the union of the coordinate axes in $\mathbf{A}^n$).*

Proof. — Let $\mathcal{J} \subset \mathcal{O}_D$ be the ideal of nilpotent elements. Filtering $\mathcal{J}$ by a chain of ideals $\mathcal{J}_k$ such that $\mathcal{J}\mathcal{J}_k \subset \mathcal{J}_{k+1}$, and using the exact sequence

$$0 \longrightarrow \mathcal{J}_k/\mathcal{J}_{k+1} \xrightarrow{a \mapsto 1+a} (\mathcal{O}_D/\mathcal{J}_{k+1})^* \longrightarrow (\mathcal{O}_D/\mathcal{J}_k)^* \longrightarrow 0,$$

it is easy to deduce that $\text{Pic}^0(D/k)$ is an extension of $\text{Pic}^0(D_{\text{red}}/k)$ by a unipotent group. Therefore, since by assumption $\text{Pic}^0(D/k)$ has no unipotent subgroups,

$$\text{Pic}^0(D/k) \simeq \text{Pic}^0(D_{\text{red}}/k).$$

Since $H^1(\mathcal{O}_D)$, resp. $H^1(\mathcal{O}_{D_{\text{red}}})$, is naturally isomorphic to the Zariski tangent space to $\text{Pic}^0(D/k)$, resp. $\text{Pic}^0(D_{\text{red}}/k)$, it follows that

$$H^1(\mathcal{O}_D) \simeq H^1(\mathcal{O}_{D_{\text{red}}}),$$

hence (i) is proven. Let $\pi : C \rightarrow D_{\text{red}}$ be the normalization of D_{red} and let D^* be the local ringed space which, as topological space is D , and whose structure sheaf is given by

$$\Gamma(U, \mathcal{O}_{D^*}) = \{f \in \Gamma(U, \pi_*(\mathcal{O}_C)) \mid x_1, x_2 \in \pi^{-1}(U), f(x_1) = f(x_2) \text{ if } \pi(x_1) = \pi(x_2)\}.$$

It is easy to check that D^* is a 1-dimensional scheme whose singularities are all transversal crossings of a set of non-singular branches and that π factors:

$$C \xrightarrow{\pi'} D^* \xrightarrow{\pi''} D_{\text{red}}.$$

I claim that $\pi'' : D^* \rightarrow D_{\text{red}}$ is an isomorphism. Filter $\mathcal{O}_{D^*}/\mathcal{O}_{D_{\text{red}}}$ so as to obtain a chain of coherent $\mathcal{O}_{D_{\text{red}}}$ -algebras

$$\mathcal{O}_{D^*} = \mathcal{O}^{(0)} \supset \mathcal{O}^{(1)} \supset \dots \supset \mathcal{O}^{(k)} = \mathcal{O}_{D_{\text{red}}}$$

such that $\ell(\mathcal{O}^{(n)}/\mathcal{O}^{(n+1)}) = 1$. Equivalently, this factors π'' :

$$D^* = D_0 \longrightarrow D_1 \longrightarrow \dots \longrightarrow D_k = D_{\text{red}},$$

where $\mathcal{O}^{(n)} \simeq \mathcal{O}_{D_n}$, and all arrows are homeomorphisms. If $\{x_n\} = \text{Supp}(\mathcal{O}^{(n)}/\mathcal{O}^{(n+1)})$, then we get an exact sequence:

$$0 \longrightarrow \mathcal{O}_{D_{n+1}}^* \longrightarrow \mathcal{O}_{D_n}^* \longrightarrow k_{x_n} \longrightarrow 0$$

(k_y denotes the residue field at y , as sheaf on D), and hence:

$$0 \longrightarrow k \longrightarrow H^1(\mathcal{O}_{D_{n+1}}^*) \longrightarrow H^1(\mathcal{O}_{D_n}^*) \longrightarrow 0.$$

It follows easily that $\text{Pic}^0(D_{n+1}/k)$ is an extension of $\text{Pic}^0(D_n/k)$ by G_a . But $\text{Pic}^0(D_{\text{red}}/k)$ has no unipotent subgroups, and this can only happen if $D_{\text{red}} = D^*$. Q.E.D. for lemma.

We apply the lemma to $\mathcal{C}_s$. According to Lichtenbaum [L] and Safarevich [S], there is a divisor K on $\mathcal{C}$ such that for all positive divisors D lying over the closed point of $\text{Spec}(R)$, we have:

$$\chi(\mathcal{O}_D) = -\frac{(D \cdot (D + K))}{2}.$$

Let $E_1, \dots, E_n$ be the components of $\mathcal{C}_s$, $d_1, \dots, d_n$ their multiplicities. Then conclusion (i) of the lemma implies that:

$$\left(\sum_i d_i E_i \cdot \left(\sum_i d_i E_i + K \right) \right) = \left(\sum_i E_i \cdot \left(\sum_i E_i + K \right) \right).$$

But $((\sum_i d_i E_i) \cdot E_k) = 0$, all k , since $\sum_i d_i E_i$ is the divisor of a function $\pi \in R$ if $(\pi) =$ maximal ideal of R . Therefore:

$$(*) \quad \left(\left(\sum_i (d_i - 1) E_i \right) \cdot K \right) = \left(\sum_i E_i \cdot \sum_i E_i \right).$$

Note that at least one d_i equals 1 since $\mathcal{C}$ has a section over $\text{Spec}(R)$ and every section must pass through a component of $\mathcal{C}_s$ of multiplicity 1. Moreover the intersection matrix $(E_i \cdot E_j)$ is negative indefinite, with one-dimensional degenerate subspace generated by $\sum_i d_i E_i$, hence if some $d_i > 1$, it follows that $(\sum_i E_i \cdot \sum_i E_i) < 0$. Therefore, by $(*)$, $(E_{i_0} \cdot K) < 0$ for some i_0 . Then we have:

- a) $(E_{i_0} \cdot K) < 0$;
- b) $(E_{i_0} \cdot E_{i_0}) < 0$;
- c) $(E_{i_0} \cdot (E_{i_0} + K)) = -2\chi(\mathcal{O}_{E_{i_0}}) \geq -2$;

hence in fact $(E_{i_0} \cdot E_{i_0}) = (E_{i_0} \cdot K) = -1$, so E_{i_0} is an exceptional curve of the first kind. This contradicts our assumption that on $\mathcal{C}$ all possible curves have been blown down. Therefore $d_i = 1$, all i .

This proves that $\mathcal{C}_s$ is reduced. By conclusion (ii) of the lemma, plus the fact that the dimension of its Zariski tangent-space is everywhere one or two (since $\mathcal{C}_s$ lies on a regular surface $\mathcal{C}$), we deduce that $\mathcal{C}_s$ has only double points. This proves that C has stable reduction in the case that $C(K) \neq \emptyset$.

In the general case, C will acquire a rational point in a finite extension K' of K . Let S' be the spectrum of the localisation at some maximal ideal of the integral closure of R in K' ; S' is the spectrum of a valuation ring and is faithfully flat over S . We put $S'' = S' \times_S S'$ and denote by C' and C'' the inverse images of C on $S'_\eta = \text{Spec}(K')$ and S''_η respectively.

$$\begin{array}{ccccc} C'' & \longrightarrow & C' & \longrightarrow & C \\ \downarrow & & \downarrow & \downarrow & \\ S'' & \xrightarrow[\text{pr}_2]{\text{pr}_1} & S' & \xrightarrow{p} & S. \end{array}$$

Let $\overline{C}'$ be the stable curve on S' having C' as generic fibre. The restriction, C' , of $\overline{C}'$ to S'_η carries a descent datum with respect to p , i.e. an isomorphism, φ_η , between the restrictions of $\text{pr}_1^*(\overline{C}')$ and $\text{pr}_2^*(\overline{C}')$ to S''_η . It remains only to extend the isomorphism φ_η to an isomorphism, φ , between the $\text{pr}_i^*(\overline{C}')$. Then φ will be a descent datum for $\overline{C}'$ with respect to p . As $\overline{C}'$ is canonically polarized (1.2), this descent datum will be effective, and so define a stable curve, $\overline{C}$, over S with generic fibre C .

Because $\mathcal{J}_s^0$ has no unipotent radical, the inverse image, $p^* \mathcal{J}^0$, of $\mathcal{J}^0$ on S' is the identity component of the Néron Model of the jacobian of C' , so that, defining $q = p \circ \text{pr}_1 = p \circ \text{pr}_2$, one has

$$\begin{aligned} \text{Pic}^0(\overline{C}'/S') &= p^* \mathcal{J}^0, \\ \text{Pic}^0(\text{pr}_1^* \overline{C}') &= \text{Pic}^0(\text{pr}_2^* \overline{C}') = q^* \mathcal{J}^0. \end{aligned}$$

We denote by T the closed subscheme of $\text{Isom}(S'', \text{pr}_1^*(\overline{C}'), \text{pr}_2^*(\overline{C}'))$ corresponding to those isomorphisms which induce (via the preceding identifications) the identity on the inverse image of $\mathcal{J}^0$.

By (1.11), T is finite and unramified over S'' , and by (1.13) T is radicial over S'' . We conclude that the morphism from T to S'' identifies T with a closed subscheme X of S'' . As X contains S''_η , which is schematically dense in S'' , we have that X is S'' and T “is” the desired section, φ , of $\text{Isom}(S'', \text{pr}_1^*(\overline{C}'), \text{pr}_2^*(\overline{C}'))$ over S'' which extends φ_η . Q.E.D.

Combining Proposition (2.3), Theorem (2.4), and the stable reduction theorem for abelian varieties quoted in the introduction, we obtain the most important consequence:

Corollary (2.7). — Let R be a discrete valuation ring with quotient field K . Let C be a smooth geometrically irreducible curve over K of genus $g \geq 2$. Then there exists a finite algebraic extension L of K and a stable curve $\mathcal{C}_L$ over R_L , the integral closure of R in L , with generic fibre $\mathcal{C}_{L,\eta} = C \times_K L$.

§3. Elementary derivation of the theorem.

Let k be an algebraically closed field of char. $p \neq 0$. We use the notation of § 1, except that we will now denote by H_g the product $H_g \times \text{Spec}(k)$ of the previous H_g with $\text{Spec}(k)$: it is a disjoint union of non-singular varieties

$H_{g,1}, \dots, H_{g,n}$ over k and is the subscheme of $\text{Hilb}_{\mathbf{P}^{5g-6}/k}^{P_g}$ of tri-canonical stable curves. Similarly, H_g^0 is the open dense subset of H_g of tri-canonical non-singular curves. By the results of [M₁], we know that a coarse geometric quotient

$$M_g^0 = H_g^0 / \text{PGL}(5g-6)$$

exists, that it is a disjoint union of normal varieties over k and is the coarse moduli space for non-singular curves of genus g . Let $S^* = H_g - H_g^0$. Then everything decomposes into the same set of components.

Let

$$\begin{aligned} H_{g,i} &= \text{components of } H_g, \\ \text{then } H_{g,i}^0 &= H_{g,i} \cap H_g^0 = \text{components of } H_g^0, \\ \text{and } M_{g,i}^0 &= H_{g,i}^0 / \text{PGL}(5g-6) = \text{components of } M_g^0, \\ \text{and } S^* &= \text{disjoint union of } S_1^*, \dots, S_n^*, \quad S_i^* = H_{g,i} \cap S^*. \end{aligned}$$

We want to prove that M_g^0 , or equivalently H_g^0 , or equivalently H_g is irreducible. We shall use: (i) the fact that these statements are true in char. 0; (ii) the inductive assumption that these statements are true for smaller genus.

Step I. — No component of M_g^0 is complete (i.e., proper over k).

Proof. — Here we use the char. 0 result. By [M₁], there is a scheme X , quasi-projective over $\text{Spec}(W(k))$, $W(k)$ the Witt vectors, whose closed fibre is M_g^0 , and whose generic fibre X_η is the char. 0 coarse moduli space over the quotient field of $W(k)$. In particular, X_η is known to be connected. Since X is quasi-projective over $W(k)$, we can embed X as an open dense subset of a scheme $\bar{X}$ projective over $W(k)$. $\bar{X}_\eta$ is still connected, hence by the connectedness theorem of Enriques-Zariski [EGA 3], the closed fibre $\bar{X}_0$ of $\bar{X}$ is connected. But if Y were a complete variety which is a component of M_g^0 , then: a) Y is an open subset of M_g^0 , which is an open subset of $\bar{X}_0$, and: b) since Y is proper/ k , Y would be a closed subset of $\bar{X}_0$ too. Therefore, $Y = \bar{X}_0$, hence M_g^0 is itself irreducible and complete. On the other hand, if A_g is the coarse moduli space of principally polarized g -dimensional abelian varieties, then the map associating to each curve its jacobian defines a morphism:

$$\theta : M_g^0 \rightarrow A_g.$$

If M_g^0 were complete, the image of θ would be closed. But it is well known that the closure of the image of θ contains all products of lower dimensional jacobians too, so it is not closed. Q.E.D.

Step II. — No component of H_g consists entirely of non-singular curves, i.e., $S_i^* \neq \emptyset$ for all i .

Proof. — Here we combine Step I with the result of § 2. Take any i . Let $T = \text{Spec } k[[t]]$. Since $M_{g,i}^0$ is not complete, there is a morphism φ of the generic point T_η of T into $M_{g,i}^0$ which does not extend to a morphism of T into $M_{g,i}^0$. Now we replace T by its normalization T' in a finite algebraic extension and let $\varphi' : T'_\eta \rightarrow M_{g,i}^0$ be the induced morphism, which still does not extend to a morphism from T' to $M_{g,i}^0$. By the results of § 2, if T' is chosen suitably there exists a stable curve $\pi : C' \rightarrow T'$ over T' whose generic fibre C'_η is a non-singular curve corresponding to the morphism $\varphi' : T'_\eta \rightarrow M_{g,i}^0$ via the functorial properties of the moduli space. Since T' is the spectrum of a local ring, we can choose an isomorphism

$$\mathbf{P}(\pi_*(\omega_{C'/T'}^{\otimes 3})) \simeq \mathbf{P}^{5g-6} \times T'$$

and get a tri-canonical embedding $C' \subset \mathbf{P}^{5g-6} \times T'$. C' , with this embedding, is then induced from the universal tri-canonically embedded stable curve by a morphism $\psi : T' \rightarrow H_g$. Since the generic fibre of C' is C'_η , we get a commutative diagram:

$$\begin{array}{ccc} T' & \xrightarrow{\psi} & H_g \\ \cup & & \cup \\ T'_\eta & \xrightarrow{\text{res}(\psi)} & H_g^0 \\ \varphi' \searrow & & \downarrow \\ & M_{g,i}^0 & \subset M_g^0. \end{array}$$

If $\text{Im}(\psi) \subset H_g^0$, then φ' would extend to a morphism from T' to M_g^0 , hence from T' to $M_{g,i}^0$, and this is a contradiction. Moreover, $\psi(T_\eta)$ must be a point of $H_{g,i}^0$, since its image in M_g^0 is in $M_{g,i}^0$. Therefore the image x of the closed point of T' by ψ is in the closure of $H_{g,i}^0$, i.e., in $H_{g,i}$, but not in $H_{g,i}^0$ itself. Q.E.D.

Step III. — S^* is connected.

Proof. — This will follow using only the induction assumption of irreducibility for lower genera. Let $Z \subset H_g \times \mathbf{P}^{5g-6}$ be the universal tri-canonically embedded stable curve. Let $S \subset Z$ be the set of points where Z is not smooth over H_g . As we proved in § 1, S is non-singular and is the normalization of S^* . In particular, this shows that if $x \in S^*$, then the corresponding curve Z_x has exactly one double point if and only if x is a non-singular point of S^* . Stable curves C of genus g with exactly one double point belong to one of the following types:

$$\begin{array}{ll} \text{type } 0 : & \textcircled{\ominus} \quad \begin{array}{l} C \text{ irreducible} \\ \text{normalization } C' \text{ of } C \text{ has genus } g-1; \end{array} \\ \text{type } k : & \begin{array}{l} \text{X} \\ \text{X} \end{array} \quad \begin{array}{l} C \text{ has two non-singular components } C_1, C_2 \\ \text{genus}(C_1) = k \\ \text{genus}(C_2) = g-k \end{array} \end{array}$$

where $1 \leq k \leq \lfloor \frac{g}{2} \rfloor$. If $0 \leq k \leq \lfloor \frac{g}{2} \rfloor$, let

$$S^*(k) = \{x \in S^* \mid Z_x \text{ has one double point and is of type } k\}.$$

Then the open dense subset of S^* of non-singular points is the disjoint union of open subsets $S^*(0), \dots, S^*(\lfloor \frac{g}{2} \rfloor)$. We first check:

(*) Each set $S^*(k)$ is irreducible.

Proof of ().* — Take the case $k = 0$: the cases $1 \leq k \leq \lfloor \frac{g}{2} \rfloor$ are similar. Let

$$T = \{\{x_1, x_2\} \mid x_1 \neq x_2 \text{ and } \pi(x_1) = \pi(x_2) \in H_{g-1}^0\} \subset Z_{g-1} \times_{H_{g-1}} Z_{g-1}.$$

T is smooth with irreducible fibres over H_{g-1}^0 , hence T is irreducible. Consider the correspondence relating $S^*(0)$ and T :

$$W = \left\{ \begin{array}{l} \text{set of pairs } x \in S^*(0), \{x_1, x_2\} \in T \text{ such that if } y = \pi(x_1) = \pi(x_2), \\ \text{then there exists a birational morphism } f : (Z_{g-1})_y \rightarrow (Z_g)_x \\ \text{such that } f(x_1) = f(x_2). \end{array} \right.$$

It is easy to check that W is Zariski-closed. Moreover, for any $\{x_1, x_2\} \in T$, $W \cap (S^*(0) \times \{x_1, x_2\})$ is an orbit in $S^*(0)$ under $\text{PGL}(5g-6)$: so these are non-empty irreducible subsets all of the same dimension. It follows that W itself is irreducible. But the projection from W to $S^*(0)$ is surjective, so $S^*(0)$ is irreducible. Q.E.D. for (*).

Now for any k , $1 \leq k \leq \lfloor \frac{g}{2} \rfloor$, choose any stable curve $C(k)$ of the type:

$$\begin{array}{ll} \begin{array}{c} \text{X} \\ \text{X} \end{array} \begin{array}{l} Q_1 \\ Q_2 \end{array} & \begin{array}{l} \text{one non-singular component } C(k)' \text{ of genus } k, \\ \text{one component } C(k)'' \text{ with one double point, normalization of genus } g-k-1. \end{array} \end{array}$$

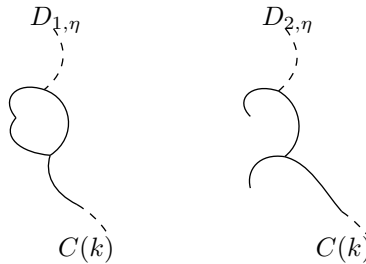
Let $P(k)$ be a point of H_g such that $Z_{P(k)} \simeq C(k)$. Step III will be completed if we prove:

(**) $P(k)$ is in the closure of $S^*(0)$ and of $S^*(k)$, hence $S^*(0), S^*(k)$ both lie in the same topological component of S^* .

*Proof of (**).* — Let $T = \text{Spec } k[[t]]$. Using the fact that S^* has two branches through $P(k)$, one for each of the double points of $C(k)$, we see that there exist two morphisms

$$f_1, f_2 : T \rightarrow S^*,$$

$$f_1(T_s) = f_2(T_s) = P(k), \quad T_s = \text{closed point of } T,$$



such that if $\pi_1 : D_1 \rightarrow T$, $\pi_2 : D_2 \rightarrow T$ are the two stable curves over T induced by f_1 and f_2 , then: a) the closed fibres $D_{1,s}$, $D_{2,s}$ are $C(k)$; b) there are sections $s_1 : T \rightarrow D_1$, $s_2 : T \rightarrow D_2$ whose images are non-smooth points of π_1 , π_2 and such that $s_1(T_s)$ and $s_2(T_s)$ are the two double points Q_1 and Q_2 of $C(k)$ respectively; and c) the generic fibres $D_{1,\eta}$, $D_{2,\eta}$ have only one double point (cf. figure). I claim:

A) $D_{1,\eta}$ is of type (k) ;

B) $D_{2,\eta}$ is of type (0) .

To prove A), let D'_1 be the result of blowing up the subscheme $s_1(T)$ of D_1 . Then D'_1 is still flat and proper over T and its special fibre is the special fibre of D_1 with Q_1 blown up (use the fact that formally at Q_1 , D_1 is isomorphic to $k[[t, x, y]]/(xy)$ with the section given by $x = y = 0$). Therefore the special fibre of D'_1 is the disjoint union of $C(k)'$, $C(k)''$, so the general fibre of D'_1 is the disjoint union of two irreducible curves which specialize to $C(k)'$, $C(k)''$ respectively. Since $D_{1,\eta}$ has only one double point, $D'_{1,\eta}$ is non-singular, so $D'_{1,\eta}$ is the disjoint union of two non-singular irreducible curves which must then have the same genera as $C(k)'$ and $C(k)''$, i.e., k , $g - k$. Thus $D_{1,\eta}$ has type (k) .

To prove B), it suffices to check that $D_{2,\eta}$ is geometrically irreducible. If not, $D_{2,\eta}$ would have two components meeting at the single point $s_2(T_\eta)$. Since D_2 is smooth over T at each generic point of its special fibre, distinct geometric components of $D_{2,\eta}$ have to have specializations which are distinct components of $(D_2)_s$. Then $(D_2)_s$ would have two components meeting at the point $Q_2 = s_2(T_s)$. This is false, so B) is proven.

Now because of A), $f_1(T_\eta) \in S^*(k)$, hence $P(k) = f_1(T_s)$ is in the closure of $S^*(k)$. And because of B), $f_2(T_\eta) \in S^*(0)$, hence $P(k) = f_2(T_s)$ is in the closure of $S^*(0)$ too. Q.E.D. for (**).

This completes the proof of Step III since we now see that all irreducible components of S^* are part of the same topological component. Finally, from Steps II and III, we see that

a) S^* , being connected, is part of a single component of H_g , while

b) each component of H_g contains part of S^* .

Thus H_g is irreducible, as was to be proven.

§4. Some results on algebraic stacks.

The proofs of the results stated in this section will be given elsewhere.

Let C be a category and let $p : \mathcal{S} \rightarrow C$ be a category over C . For each $U \in \text{Ob } C$, we denote by $\mathcal{S}_U$ the fibre $p^{-1}(U)$. The category $\mathcal{S}$ is *fibered in groupoids* over C if the following two conditions are verified:

a) For all $\varphi : U \rightarrow V$ in C and $y \in \text{Ob } \mathcal{S}_V$ there is a map $f : x \rightarrow y$ in $\mathcal{S}$ with $p(f) = \varphi$.

b) Given a diagram

$$\begin{array}{ccc} & x & \\ & \searrow f & \\ y & \xrightarrow{\quad g \quad} & z \end{array}$$

in $\mathcal{S}$, let

$$\begin{array}{ccc} & U & \\ & \searrow \varphi & \\ V & \xrightarrow{\quad \psi \quad} & W \end{array}$$

be its image in C . Then for all $\chi : U \rightarrow V$ such that $\varphi = \psi\chi$, there is a unique $h : x \rightarrow y$ such that $f = g \cdot h$ and $p(h) = \chi$.

Condition b) implies that the $f : x \rightarrow y$ whose existence is asserted in a) is unique up to canonical isomorphism.

Assume that for each $\varphi : U \rightarrow V$ in C and each $Y \in \text{Ob } \mathcal{S}_V$, such an $f : x \rightarrow y$ has been chosen. This x will be written as φ^*y . Then, φ^* “is” a functor from $\mathcal{S}_V$ to $\mathcal{S}_U$ and if $\varphi\psi$ is a composite morphism in C , the functors $(\varphi\psi)^*$ and $\psi^*\varphi^*$ are canonically isomorphic.

We propose the terminology “stack” for the French word “champ” of non-abelian cohomology (Giraud [G]).

Definition (4.1). — Let C be a category with a Grothendieck topology. We assume products and fibre products exist in C . A *stack in groupoids* over C is a category over C , $p : \mathcal{S} \rightarrow C$ such that:

(i) $\mathcal{S}$ is fibered in groupoids over C .

(ii) For any $U \in \text{Ob } C$ and any objects x, y in $\mathcal{S}_U$ the functor from C/U to (sets) which to any $\varphi : V \rightarrow U$ associates $\text{Hom}_{\mathcal{S}_V}(\varphi^*x, \varphi^*y)$ is a sheaf.

(iii) If $\varphi_i : V_i \rightarrow U$ is a covering family in C , any descent datum relative to the φ_i , for objects in $\mathcal{S}$, is effective.

For each $x \in \text{Ob } \mathcal{S}_U$, there are given isomorphisms between the inverse images of $x_i = \varphi_i^*x$ and $x_j = \varphi_j^*x$ over $V_{ij} = V_i \times_U V_j$, and the pull-backs of these isomorphisms on $V_{ijk} = V_i \times_U V_j \times_U V_k$ satisfy a “cocycle” condition. In (iii) it is required that reciprocally, any such “descent datum” be defined by some $x \in \mathcal{S}_U$.

In what follows, for the sake of brevity, we will use “stack” to mean “stack in groupoids”.

If $U \in \text{Ob } C$ and if $\mathcal{S}$ is a stack over C , the fibre $\mathcal{S}_U$ will be called the category of sections of $\mathcal{S}$ over U .

Let C be as in (4.1). The stacks over C are the objects of a 2-category [B] (stacks/ C): 1-morphisms are functors from one stack to another, compatible with the projection into C ; and 2-morphisms are morphisms of functors. In this 2-category every 2-morphism is an isomorphism. Products and 2-fibre products exist in this 2-category.

To each $X \in \text{Ob } C$ is associated the “representable” stack over C whose category of sections over U is the discrete category whose objects are the morphisms from U to X . This stack will be denoted simply X . For any stack $\mathcal{S}$, the category

$$\text{Hom}(X, \mathcal{S})$$

is canonically equivalent to the category of sections of $\mathcal{S}$ over X . Because of this, $\mathcal{S}$ is sometimes said to “classify” its sections over variable $X \in \text{Ob } C$. In the category $\text{Hom}(\mathcal{S}, X)$, all morphisms are identities, i.e., $\text{Hom}(\mathcal{S}, X)$ is just a set.

Let us denote also by C the 2-category having the same objects and morphisms as C , and in which the identities are the only 2-morphisms. The above construction then identifies C with a full sub-2-category of (stacks/ C).

For each $S \in \text{Ob } C$, the category C/S satisfies the assumptions of (4.1). Any stack $\mathcal{S}_0$ over C/S (an “ S -stack”) gives rise to a stack $\mathcal{S}$ over C ; a section (φ, η) of $\mathcal{S}$ over $U \in \text{Ob } C$ consists of

- (i) a morphism $\varphi : U \rightarrow S$;
- (ii) a section η of $\mathcal{S}_0$ over (U, φ) .

Definition (4.2). — A 1-morphism of stacks over C , $F : \mathcal{S}_1 \rightarrow \mathcal{S}_2$, will be called *representable* if for any X in C and any 1-morphism $x : X \rightarrow \mathcal{S}_2$, the fibre product $X \times_{\mathcal{S}_2} \mathcal{S}_1$ is a representable stack.

In down to earth terms, this means the following:

- (i) for any $f : Y \rightarrow X$ in C , the category whose objects are pairs

$$\{\text{a section, } y, \text{ of } \mathcal{S}_1 \text{ over } Y; \text{ an isomorphism } F(y) \xrightarrow{\sim} f^*(x)\}$$

is equivalent to a category $S(f)$ in which all morphisms are identities;

- (ii) the functor $f \mapsto \text{Ob } S(f)$ is representable by some $g : Z \rightarrow X$. Such a Z represents the fibre product $X \times_{\mathcal{S}_2} \mathcal{S}_1$.

Let P be a property of morphisms in C , stable by change of base and of a local nature on the target.

Definition (4.3). — A representable morphism $F : \mathcal{S}_1 \rightarrow \mathcal{S}_2$ of stacks over C has property P if for any 1-morphism $x : X \rightarrow \mathcal{S}_2$ the morphism in C deduced by base change: $F' : X \times_{\mathcal{S}_2} \mathcal{S}_1 \rightarrow X$ has that property.

Proposition (4.4). — Let $\mathcal{S}$ be a stack. The diagonal map

$$\mathcal{S} \rightarrow \mathcal{S} \times \mathcal{S}$$

is representable if and only if for all $X, Y \in \text{Ob } C$ and 1-morphisms $x : X \rightarrow \mathcal{S}$, $y : Y \rightarrow \mathcal{S}$, the fibre product $X \times_{\mathcal{S}} Y$ is representable.

If $X \in \text{Ob } C$ and x, y are sections of $\mathcal{S}$ over X , we denote by $\text{Isom}(X, x, y)$ the sheaf on C/X which to every Z over X associates the set of isomorphisms between the inverse images of x and y over Z . Then the object representing $\mathcal{S} \times_{(\mathcal{S} \times \mathcal{S})} X$ (the product taken with the map $(x, y) : X \rightarrow \mathcal{S} \times \mathcal{S}$) is just $\text{Isom}(X, x, y)$. If $X \rightarrow \mathcal{S}, Y \rightarrow \mathcal{S}$ are 1-morphisms, then the object representing $X \times_{\mathcal{S}} Y$ is just $\text{Isom}(X \times Y, p_1^*x, p_2^*y)$.

Henceforth, C will be the category of schemes with the étale topology (SGAD, IV, (6.3)).

Definition (4.5). — A stack $\mathcal{S}$ is *quasi-separated* if the diagonal morphism from $\mathcal{S}$ to $\mathcal{S} \times \mathcal{S}$ is representable, quasi-compact and separated.

Definition (4.6). — A stack $\mathcal{S}$ is an *algebraic stack*¹ if

¹This definition is the “right” one only for quasi-separated stacks. It will however be sufficient for our purposes.

- (i) $\mathcal{S} \rightarrow \mathcal{S} \times \mathcal{S}$ is representable;
- (ii) there exists a 1-morphism $x : X \rightarrow \mathcal{S}$ such that for all $y : Y \rightarrow \mathcal{S}$, the projection morphism $X \times_{\mathcal{S}} Y \rightarrow Y$ is surjective and étale (i.e., x is étale and surjective).

The 2-category of algebraic stacks contains the representable stacks and is stable under products and fibre products. If $\mathcal{S}$ is a quasi-separated algebraic stack, the diagonal map is unramified and quasi-affine.

Definition (4.7). — An algebraic stack $\mathcal{S}$ is separated if $\mathcal{S} \rightarrow \mathcal{S} \times \mathcal{S}$ is proper (or, equivalently, finite). A 1-morphism $f : \mathcal{S}_1 \rightarrow \mathcal{S}_2$ is separated (resp. quasi-separated) if for any morphism $x : X \rightarrow \mathcal{S}_2$ from a separated scheme X to $\mathcal{S}_2$, the fibre product $\mathcal{S}_1 \times_{\mathcal{S}_2} X$ is separated (resp. quasi-separated).

Example (4.8). — Let X be a scheme over S . Let G be a group scheme over S , étale, separated and of finite type over S , which operates on X . We will denote by $[X/G]$ the S -stack whose category of sections over an S -scheme T is the category of principal homogeneous spaces (p.h.s.) E over T , with structural group G (i.e., a p.h.s. under G_T), provided with a G -morphism $\varphi : E \rightarrow X$. The principal homogeneous space $G \times X$ over X (G acting only on the first factor) plus the G -morphism $G \times X \rightarrow X$ (given by the action of G on X) is a section of $[X/G]$ over X . The corresponding morphism $q : X \rightarrow [X/G]$ is étale and surjective, so that $[X/G]$ is an algebraic stack. In addition, X is a principal homogeneous space over $[X/G]$; the stack $[X/G]$ is representable if and only if X is a principal homogeneous space over a scheme Y , in which case

$$[X/G] \sim Y.$$

If $X = S$, then $[X/G] = [S/G]$ might be called the “classifying stack” of G over S .

Example (4.9). — Suppose a stack $\mathcal{S}$ has the property that in each category $\mathcal{S}_X$ the only morphisms are the identity morphisms. Then $\mathcal{S}_X$ is just a set $\mathcal{F}(X) = \text{Ob } \mathcal{S}_X$, and this set, under pull-back, is a contravariant functor $\mathcal{F}$ in X . Conditions (ii) and (iii) of (4.1) assert that the functor $\mathcal{F}$ is a sheaf on C . Artin and Knutson [K] have defined an *algebraic space* to be a sheaf $\mathcal{F}$ such that:

- (i) for any morphisms $X \rightarrow \mathcal{F}, Y \rightarrow \mathcal{F}$ of representable functors to $\mathcal{F}$, the fibre product $X \times_{\mathcal{F}} Y$ is representable;
- (ii) there exists a morphism $X \rightarrow \mathcal{F}$, represented by surjective, étale morphisms of schemes.

This is exactly what we have called an algebraic stack in this case.

Definition (4.10). — Let $\mathcal{S}$ be an algebraic stack. The étale site $\mathcal{S}_{et}$ of $\mathcal{S}$ is the category with objects the étale morphisms

$$x : X \rightarrow \mathcal{S}$$

and where a morphism from (X, x) to (Y, y) is a morphism of schemes $f : X \rightarrow Y$ plus a 2-morphism between the 1-morphism $x : X \rightarrow \mathcal{S}$ and $y \circ f : X \rightarrow \mathcal{S}$. A collection of morphisms $f_i : (X_i, x_i) \rightarrow (X, x)$ is a covering family if the underlying family of morphisms of schemes is surjective.

The site $\mathcal{S}_{et}$ is in a natural way ringed. When we speak of sheaves on $\mathcal{S}$ we mean sheaves on $\mathcal{S}_{et}$.